

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I H MOHAMED FAAIZ 192524099 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Scenario Based

1. Cloud architecture using scalability and caching

- * Deploy the video streaming platform on multiple cloud servers across different regions

- * Use a content delivery network (CDN) to cache video closer to user reducing buffering and latency

- * Use distributed data bases and cloud networking for reliable data access

Local Balancing and Auto scaling

- * Local Balancing Distributes incoming user requests evenly across multiple servers preventing server overload

- * Auto scaling :: Automatically adds new servers when user traffic increases and removes extra servers when traffic decreases

Cost optimization strategies

- * Use a pay as you go pricing model to pay only for resources used
- * Enable auto scaling to avoid paying for idle servers during traffic
- * Store frequently accessed videos in CDN cache and archive rarely video in low cost cloud storage

27 Fintech Startup - Multi cloud Containers

Soln:-

- Use docker containers
- Use Kubernetes for containers orchestration
- Deploy apps on multiple cloud providers

Service Discovery and Scaling

- Kubernetes automatically find services
- It increase or decrease containers based on demand

Benefits:-

- High durability
- Better Reliability
- Avoids Render lock-in

Risks:-

- More complex Managements
- Higher network Cost
- Security management difficult

3) Elearning platforms - Backup and Disaster Recovery

Solution:-

- Regular used backups
- Enable data replication
- Use versioning to recover deleted files

RTO and RPO:-

- RTO: Restore the system quickly
- RPO: Restore the latest available data with minimum loss

Business continuity steps:-

- Keep backup in different regions
- Test Recovery regularly
- Use automatic failover

4) Government Agency - Flybrid cloud

Flybrid Architecture:-

- keep sensitive data on-premise
- Store non-sensitive data in the public cloud
- Connect both using Secure VPN

Identity management and Encryption

- Use Multi-factor Authentication (MFA)
- Encrypt data during storage and transfer
- Gives user role based access

Challenges

- Complex management
- Security and compliance issues
- Network latency

Benefits:-

- High durability
- Better Reliability
- Avoids Reader lock-in

Risks:-

- More complex Managements
- Higher network Cost
- Security management difficult

3) Eliminating platform - Backup and Disaster Recovery

Solution:-

- Regular used backups
- Enable data replication
- Use versioning to recover deleted files

RTO and RPO:-

- RTO: Restore the system quickly
- RPO: Restore the latest available data with minimum loss

57 SaaS provider - High Availability

→ Deploy the apps in Multiple Cloud regions

→ If one region fails another region continues the services

Data Replication and failures

→ Copy data to all regions

→ Use automatic failover to switch users to healthy region

Monitoring and Alerting:-

→ Use Cloud monitoring tools to detect issues early

→ Monitor server health continuously